

EE7207 Neural Networks & Deep Learning

Comprehensive CNN: From Foundations to Advanced Tricks

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1. Why CNN? Principles of Efficiency

- **MLP Limitation:** A $200 \times 200 \times 3$ image has $120k$ inputs. One FC layer with 1000 hidden units = 1.2×10^8 parameters. **Result:** High memory usage & Overfitting.
- **CNN Core Pillars:**
 - **Local Connectivity:** Each neuron looks at a small window (*Receptive Field*).
 - **Weight Sharing:** Same filter slides across the image. Feature *A* at $(0, 0)$ is detected the same way at $(10, 10)$.
 - **3D Topology:** Activations are volumes (H, W, C).

Key Term: Translation Equivariance

Weight sharing ensures that if an object moves in the image, its activation map moves accordingly.

2. The Convolutional Layer: Mechanics

- **Hyperparameters:** Filter size (f), Stride (s), Padding (p).
- **Padding:**
 - *Valid:* No padding, output shrinks.
 - *Same:* $p = (f - 1)/2$, output size = input size (if $s = 1$).
- **Filter Depth:** A filter must always have the **same depth** as the input volume (e.g., for RGB, filter is $f \times f \times 3$).

Parameter Calculation Formula

Number of params per layer =
 $[(f \times f \times \text{Input_Depth}) + 1] \times \text{Number_of_Filters}.$

3. Pooling: Reducing Complexity

- **Max Pooling:** Selects the maximum value.
 - *Pro:* Provides **Invariance** to small translations and reduces noise.
- **Average Pooling:** Computes the mean. Usually used at the **Global** level before the final Softmax.
- **Characteristics:**
 - No learnable parameters.
 - Operates on each channel **independently**.

4. Batch Normalization: The Training Accelerator

Problem: *Internal Covariate Shift* – distributions of layer inputs change during training, slowing down convergence.

- **Operation:** Normalizes activations to mean=0, var=1, then scales and shifts.

$$\hat{x} = \frac{x - E[x]}{\sqrt{\text{Var}[x] + \epsilon}}, \quad Y = \gamma \hat{x} + \beta$$

- **Why it works:** Allows higher learning rates and reduces sensitivity to **Weight Initialization**.
- **The (C,) Shape Logic:** Because of weight sharing, we want the same normalization for all spatial locations in a channel to maintain feature consistency.

5. Classic Architectures: ResNet & MobileNet

- **ResNet (Lecture 6):** Identity Shortcut.
 - Mitigates **Vanishing Gradient**.
 - Gradient can "skip" layers during backprop via $H(x) = F(x) + x$.
- **MobileNet (Lecture 12): Depthwise Separable Conv.**
 - Splits standard conv into:
 - 1 **Depthwise:** $f \times f \times 1$ per channel.
 - 2 **Pointwise:** $1 \times 1 \times C$ to combine channels.
 - **Efficiency:** Reduces computation by $\sim 8\times$ to $9\times$.

6. Training Tricks (Lecture 12)

- **Weight Initialization:**

- **He (MSRA) Initialization:** Best for **ReLU** ($Var = 2/n$).
- **Xavier:** Best for **Tanh/Sigmoid** ($Var = 1/n$).

- **Regularization:**

- **Dropout:** Randomly zero out neurons (train only).
- **Weight Decay (L_2):** Keeps weights small to prevent overfitting.

7. Transfer Learning Strategy

Scenario	Data & Similarity	Strategy
Case I	Small + Similar	Linear Classifier (on top)
Case II	Large + Similar	Fine-tune full model
Case III	Small + Different	Classifier on earlier layers
Case IV	Large + Different	Fine-tune or Train from scratch

Heuristic: Use earlier layers for different tasks because they detect "low-level" features (edges, blobs) common to all images.

4. You are tasked with designing a deep learning model to classify rare cancer subtypes from medical images. The dataset comprises 1,000 annotated images with a significant class imbalance (approximately 10:1). The final model will be deployed on mobile devices. Answer the following questions:
- (a) Propose three effective data augmentation strategies (beyond simple rotations and flips) that are suitable for the task. For each strategy, briefly explain:
 - (i) How it generates more diverse training samples.
 - (ii) Why it is expected to mitigate the effects of limited data and class imbalance.

(6 Marks)

Scenario: Rare cancer (1,000 images), 10:1 imbalance.

- **Mixup:** Mix two images and their labels linearly.
 - *Logic:* Smoothens decision boundaries.
- **CutMix:** Cut a patch from image B and paste onto A.
 - *Logic:* Prevents model from relying on a single visual cue.
- **GANs:** Generate minority class samples.
 - *Logic:* Balances the dataset to 1:1.

Design Choice

For small medical data, choose **CNN** over ViT due to **Inductive Bias** (Locality/Stationarity).

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(b) Consider a convolutional layer that outputs a feature map with dimensions $(N, C, H, W) = (32, 256, 28, 28)$, where 32 is the batch size, 256 is the number of channels, and 28×28 are the spatial dimensions. When applying batch normalization on this convolutional output, what is the shape of the running mean used during training? Select the correct answer from the following choices and provide a concise justification for your selection.

- (i) (256,)
- (ii) (32, 256)
- (iii) (256, 28, 28)
- (iv) (32, 28, 28)

(5 Marks)

Question

Output dim = (32, 256, 28, 28). What is the shape of **running mean**?

Correct Choice: (i) (256,)

- **Justification:** BN in CNN is **Channel-wise**. It averages over N, H, W .
- For each of the 256 channels, we store exactly **one** mean value.